A Physics-Aware Finite-Volume Study of Two-Dimen

Model formulation, conservative discretization, and verification

Abstract

This paper documents and evaluates the numerical workflow implemented in the open-source physics-aware-reservoir-agent project. The software solves steady, single-phase, incompressible Darcy flow with spatially variable, cellwise permeability on a two-dimensional rectangular domain. It uses a cell-centred finite-volume method and a two-point flux approximation (TPFA), with harmonic averaging at internal material interfaces and half-cell distances at prescribed-pressure boundaries. The pressure system is assembled as a sparse symmetric positive-definite matrix and solved with SciPy. Verification is performed using discrete conservation, global flow balance, a discrete maximum principle, exact constant-permeability and layered-medium solutions, an automated test suite, and five reference cases. In the baseline 100 × 20 homogeneous case, numerical and analytical flow rates both equal 2.0×10^{-4} m²/s per unit depth; the maximum pressure difference from the exact profile is 6.706×10^{-7} Pa, and all five applicable checks pass. Across the five reference cases, all 21 applicable checks pass at relative tolerance 10^{-9}. The results support implementation-level consistency for the stated model and test set; they do not establish field-scale predictive validity.

Keywords: Darcy flow; porous media; finite volume; TPFA; harmonic averaging; verification; mass conservation

1. Scope and research questions

The project joins a numerical PDE solver with physics-based validation utilities and an optional local language-model example. This study concerns the deterministic Darcy solver and its verification; the optional Ollama language model was not used in the numerical experiments.

- RQ1: Does the cell-centred TPFA solver reproduce the expected pressure profile and total flow for a homogeneous reservoir with fixed left/right pressures?
- RQ2: Are local cell balance, whole-domain flow balance, boundary impermeability, and pressure bounds satisfied to floating-point accuracy?
- RQ3: Does harmonic face permeability recover analytical flow through media layered in series or parallel?
- RQ4: Do these properties persist across heterogeneous, deterministic-random, and degenerate one-cell reference cases?

Study contribution

The contribution is a reproducible implementation-level assessment: equations, boundary treatment, discrete face laws, software settings, numerical outputs, and independent checks are described together. It is not a new continuum model or a comparison against laboratory or field observations.

2. Continuum model and analytical references

Let Ω = [0,L_x] × [0,L_y] denote a two-dimensional cross-section. The fluid is assumed to be single phase, incompressible, and at steady state. Permeability k(x,y) is positive and scalar; viscosity μ is constant and positive. The model omits gravity, sources/sinks, capillarity, compressibility, and time dependence.

2.1 Governing equations

Darcy's law relates the Darcy (superficial) velocity u to the pressure gradient. Combining it with steady incompressible mass conservation gives the elliptic pressure equation:

$$\mathbf{u} = -\frac{k}{\mu}\nabla p \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0 \quad \Longrightarrow \quad -\nabla \cdot \left(\frac{k}{\mu}\nabla p\right) = 0 \tag{2}$$

Because μ is spatially constant and nonzero, multiplication by μ yields the equivalent form used for the pressure system in the project:

$$-\nabla \cdot (k(x,y)\nabla p) = 0 \tag{3}$$

2.2 Boundary conditions and exact homogeneous solution

$$p(0,y) = p_L, \qquad p(L_x,y) = p_R \tag{4}$$

$$\mathbf{u} \cdot \mathbf{n} = 0 \quad \text{on } y = 0, L_y \tag{5}$$

The left and right sides prescribe pressure (Dirichlet conditions). The top and bottom are impermeable (no-flow Neumann conditions). For constant k, the exact pressure is linear in x and independent of y:

$$p(x) = p_L + \frac{p_R - p_L}{L_x}x \tag{6}$$

For a uniform medium, integration of Darcy's law over the cross-section gives total flow per unit out-of-plane depth Q = (k/μ)(pL−pR)L_y/L_x. Here Q is called "flow rate per unit depth" in the software; an actual three-dimensional rate requires specifying the omitted thickness.

2.3 Layered-media reference solutions

$$Q_{\text{series}} = \frac{L_y(p_L - p_R)}{\mu \sum_i \Delta x_i / k_i} \tag{7}$$

$$Q_{\text{parallel}} = \frac{(p_L - p_R)\sum_j k_j \Delta y_j}{\mu L_x} \tag{8}$$

Series layers place hydraulic resistances in sequence; parallel layers conduct independently and their rates add. These relations provide independent numerical references for the two principal orientations of layered permeability.

3. Numerical method: cell-centred TPFA

The rectangle is partitioned into nx × ny uniform rectangular cells, with $\Delta x = L_x/nx$ and $\Delta y = L_y/ny$. Pressure $p_{j,i}$ and permeability $k_{j,i}$ are stored at cell centres. Permeability is piecewise constant within each cell. The implementation uses one scalar permeability per cell and assumes orthogonal grid geometry.

3.1 Integrated balance and face flux

Integrating steady mass conservation over cell (j,i) states that its net outward flux is zero. For an interior face, TPFA approximates the pressure gradient using the pressure difference between adjacent cell centres; the flux from left cell L to right cell R is:

$$F_{L\to R} = \frac{T_f}{\mu}(p_L^{(c)} - p_R^{(c)}), \qquad T_f = \frac{A_f k_f}{d_{LR}} \tag{9}$$

Here A_f is face length per unit depth and dLR is centre-to-centre distance. For a face between two piecewise-constant materials, the two half-cell resistances act in series. Equating the face flux through both half-cells gives the harmonic face permeability:

$$k_f = \left[\tfrac{1}{2}\left(\frac{1}{k_L} + \frac{1}{k_R}\right)\right]^{-1} = \frac{2k_L k_R}{k_L + k_R} \tag{10}$$

3.2 Transmissibilities on this rectangular grid

$$T^x_{j,\,i+1/2} = \frac{\Delta y}{\Delta x}\,\frac{2k_{j,i}k_{j,i+1}}{k_{j,i}+k_{j,i+1}} \tag{11}$$

$$T^y_{j+1/2,\,i} = \frac{\Delta x}{\Delta y}\,\frac{2k_{j,i}k_{j+1,i}}{k_{j,i}+k_{j+1,i}} \tag{12}$$

At a prescribed-pressure boundary the distance from the cell centre to the boundary is Δx/2. Thus the left and right x-boundary transmissibility is 2kΔy/Δx. At the top and bottom, no-flow faces have zero transmissibility, so their flux is exactly zero.

3.3 Sparse linear system and reported fluxes

Summing each cell's incident transmissibilities gives the diagonal of A; each interior neighbour contributes a negative off-diagonal entry. For a cell adjacent to the left or right pressure boundary, the right-hand side receives Tboundary pboundary. The resulting system is A p=b. A is assembled in SciPy sparse CSR form and solved after conversion to CSC using spsolve. Since μ multiplies every flux uniformly, it is omitted from this pressure system and applied when fluxes are recovered.

$$A_{cc} = \sum_{f\in\partial c} T_f, \quad A_{cd} = -T_{cd}\ (d \text{ neighbour}), \quad b_c = \sum_{f\in\partial c_D} T_f p_{D,f} \tag{13}$$

The implementation computes reported face fluxes using the same transmissibilities as matrix assembly, then divides by μ. Consequently, the cell flux residual is the sparse-system residual scaled by 1/μ, up to floating-point roundoff. This consistency is the mechanism behind discrete local conservation.

4. Settings, workflow, and verification protocol

4.1 Baseline experiment

Parameter	Baseline value
Grid	nx=100, ny=20 (2,000 cells)
Domain	L_x=1,000 m; L_y=200 m
Cell dimensions	Δx=10 m; Δy=10 m
Permeability	k=1.0 × 10^{-13} m², constant
Viscosity	μ=1.0 × 10^{-3} Pa·s
Boundary pressures	pL=2.0 × 10^{7} Pa (200 bar); pR=1.0 × 10^{7} Pa (100 bar)
Boundary type	Fixed pressure left/right; no flow top/bottom
Output	outputs/baseline.png; headless plotting (Agg backend)

4.2 Reference-case suite

The validation example uses nx=40, ny=20, L_x=400 m, L_y=200 m (Δx=Δy=10 m), pL=2×10^{7} Pa, pR=10^{7} Pa, and μ=10^{-3} Pa·s for its first four cases. The single-cell case intentionally uses unit dimensions and k=1. Randomness is reproducible with NumPy seed 0.

- Homogeneous: k=10^{-13} m² everywhere.
- Series layers: four x-oriented blocks with k=[10^{-13}, 10^{-15}, 5×10^{-13}, 2×10^{-14}] m².
- Parallel layers: the same four permeability values in y-oriented blocks.
- Random log-normal: k=10^{-13} exp(2Z) m², with independent standard-normal Z generated from seed 0.
- Single cell: nx=ny=L_x=L_y=k=1; same boundary pressures and viscosity as the script.

4.3 Verification metrics and software tests

The report tolerance is rtol=10^{-9}. Relative flow errors are normalized by the largest face flux for conservation checks; pressure overshoot is normalized by a pressure scale; exact-reference checks normalize by the reference rate or boundary-pressure scale. “Local mass error” is the maximum absolute cell residual divided by the maximum absolute face flux. The automated test suite additionally checks invalid inputs, flux consistency, symmetry under pressure reversal, viscosity scaling, and whether deliberate corruption is detected by validation checks.

The project was executed in its existing Python virtual environment. Installed numerical package versions recorded for this report: NumPy 2.5.4, SciPy 1.18.1, and Matplotlib 3.11.2. The test command completed with 52 passed tests. The optional local LLM example was outside the experiment.

5. Results

5.1 Baseline homogeneous result

For the baseline settings, the exact solution predicts a uniform x-directed flow and linear cell-centre pressure. The simulation reported maximum $|p - p_{exact}| = 6.706 \times 10^{-7}$ Pa. Numerical inflow was 2.000000×10^{-4} m²/s per unit depth, matching the analytical 2.000000×10^{-4} m²/s per unit depth. The solver reported five of five checks passed; the largest baseline relative check error was 9.124×10^{-12} (global balance).

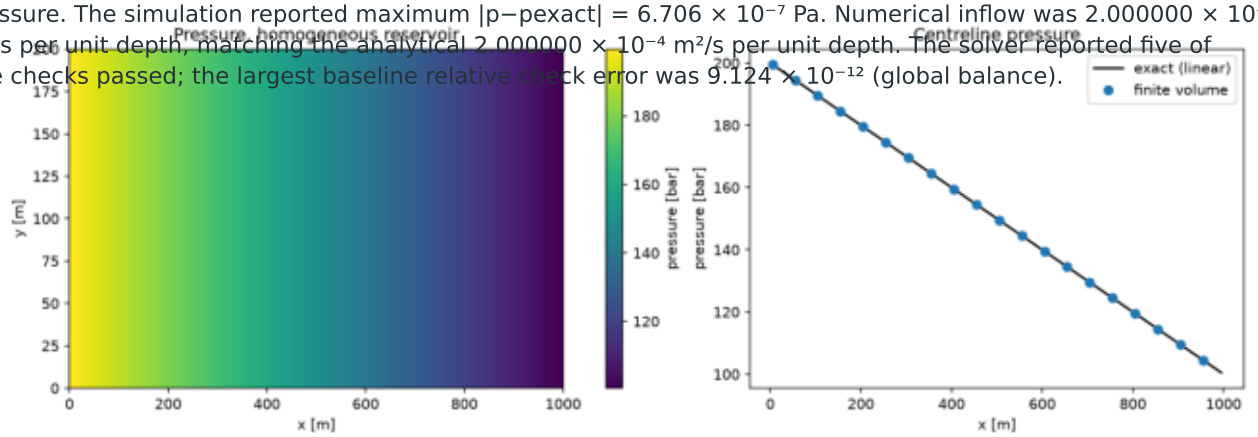


Figure 1. Baseline plot generated by examples/run_baseline.py.

5.2 Reference-suite results

Case	Checks passed	Inflow Q (m²/s per depth)	Layered/exact-flow rel. error
Homogeneous	5/5	5.000000e-04	7.823e-15
Series layers (x)	4/4	1.883239e-05	1.328e-13
Parallel layers (y)	4/4	7.762500e-04	9.288e-14
Random log-normal	3/3	3.183861e-04	2.905e-13
Single cell	5/5	1.000000e+10	0

Note: the single-cell case uses unit geometry and permeability; its very large numerical Q follows from those arbitrary unit values and is not comparable to the SI-scaled reservoir cases. The “exact-flow” error is shown only when the applicable layered/reference check exists; the single-cell value is zero because its one-cell series solution is exact.

5.3 Interpretation

All 21 applicable checks across the five reference cases passed at tolerance 10^{-9}. Local and global conservation errors were between zero and 1.088×10^{-11} relative. Pressure remained within imposed boundary values in every case. The tests demonstrate that harmonic averaging reproduces the discrete layered-flow references and that heterogeneous cases preserve conservation and the maximum principle. They test consistency of this discretization and implementation; they are not measurements of physical-model error.

6. Discussion, limitations, and reproducibility

6.1 Answers to the research questions

- RQ1: Yes for the tested homogeneous case. The discrete pressure matched the exact linear profile to 6.706×10^{-7} Pa maximum absolute error, and the flow rate matched its analytical value to the printed precision.
- RQ2: Yes for the tested cases. Local conservation, global balance, impermeable y-boundaries, and the maximum principle passed at the specified tolerance.
- RQ3: Yes for both layer orientations. Series and parallel flow matched their resistance/additive analytical formulas to relative errors 1.328×10^{-13} and 9.288×10^{-14} in the dedicated validation cases.
- RQ4: Yes within the reference suite. The random log-normal field and single-cell case passed every applicable check; the automated tests also exercise additional random fields and perturbed outputs.

6.2 Limitations and careful interpretation

- The model is steady, single-phase, incompressible, source-free, and two-dimensional. It excludes gravity, capillarity, multiphase effects, compressibility, wells, and time evolution.
- Permeability is scalar, strictly positive, and piecewise constant on a uniform orthogonal grid. TPFA can be inaccurate for strongly anisotropic tensors or non-orthogonal grids; those cases are not represented here.
- The heterogeneous field in the reference suite is a synthetic independent log-normal realization, not a measured geological sample. The separate heterogeneous example creates a correlated channel field, but that example was not used for the reported validation-table values.
- Verification against equations checks internal numerical consistency, not parameter accuracy, uncertainty, calibration, or agreement with field observations.
- The optional gpt-oss:20b/Ollama integration is an adjacent project capability, not part of the solver algorithm or the results in this paper.

6.3 Reproducibility

From the repository root, run: .venv/bin/python examples/run_baseline.py; .venv/bin/python -c 'import runpy; runpy.run_path("examples/validate_physics.py", run_name="__main__")'; and .venv/bin/python -m pytest tests/ -v. In this execution environment, direct invocation of the validation script did not resolve NumPy from its script path, so the equivalent runpy invocation was used and completed with “ALL CHECKS PASSED.” The automated test suite reported 52 passed. The baseline image is outputs/baseline.png.

References and software record

Darcy, H. (1856). Les fontaines publiques de la ville de Dijon. Victor Dalmont, Paris. Governing equations, discretization, boundary conditions, and test definitions in this paper are documented in the project files: src/reservoir_agent/darcy_solver.py; src/reservoir_agent/physics_validation.py; examples/run_baseline.py; examples/validate_physics.py; tests/test_darcy_solver.py; and tests/test_physics_validation.py. Project metadata identifies reservoir-agent version 0.1.0 and requires Python ≥3.10. Numerical-library versions for this execution are listed in Section 4.3.

Manuscript status: reproducibility-focused technical paper prepared from the repository implementation and observed runs; not peer reviewed